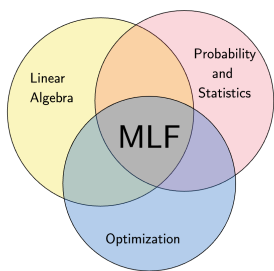


Week-7



Karthik Thiagarajan

Disclaimer: These notes were generated as a part of these two sessions:

- [session-1](#)
- [session-2](#).

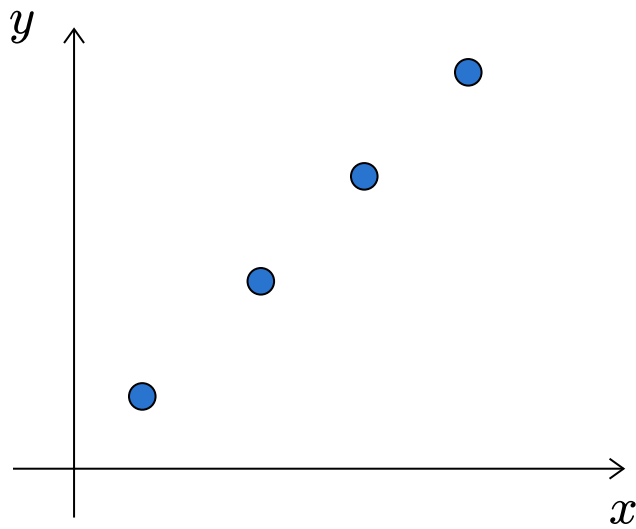
It can't be used as a standalone resource or a set of notes for this week.

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1. Dimensionality reduction

Consider a particle moving in 2D plane. We are measuring the x-coordinate and y-coordinate of the particle at different instants of time using two sensors.



Each point has a description of the dynamics of the system and we need three numbers for this:

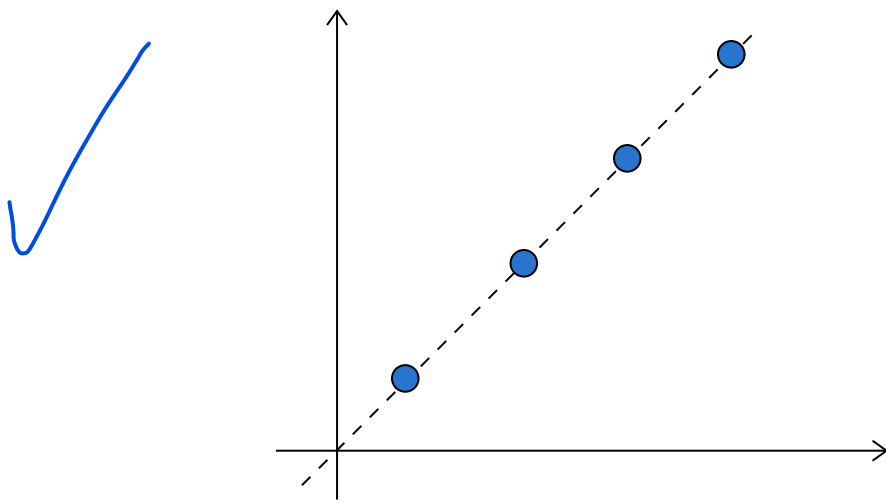
$$\underline{(t, x_t, y_t)}$$

If we have 10 measurements, it would look something like this:

$$X = \begin{bmatrix} t & 1 & 2 & \dots & 10 \\ x_t & 0.1 & 0.5 & \dots & 5.3 \\ y_t & 0.2 & 0.6 & \dots & 6.7 \end{bmatrix} \quad 3 \times 10$$

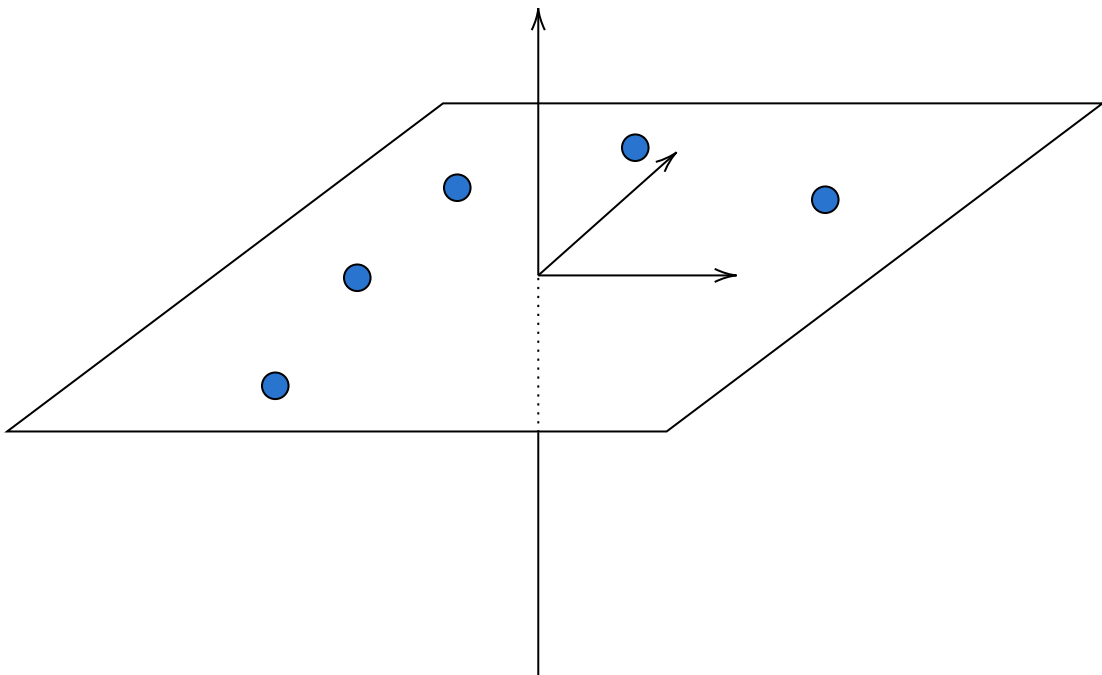
What we have is a dataset represented as a data-matrix of shape 3×10 . Each column is a data-point and each row is a feature. A feature describes a point. It is some attribute of the data-point in question. We call this representation of the dataset a data-matrix (design matrix).

Observe that the particle seems to move along a straight line and this suggests that we need only two coordinates, including time, to represent the particle's position.

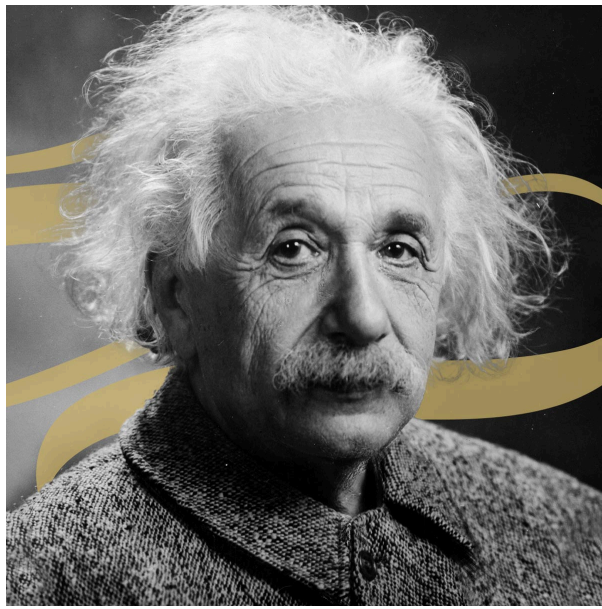


There is some pattern in the data that allows us to represent the same information with fewer dimensions. Extending this idea to higher dimensions. Consider the motion of a particle in 3D space. This requires four coordinates:

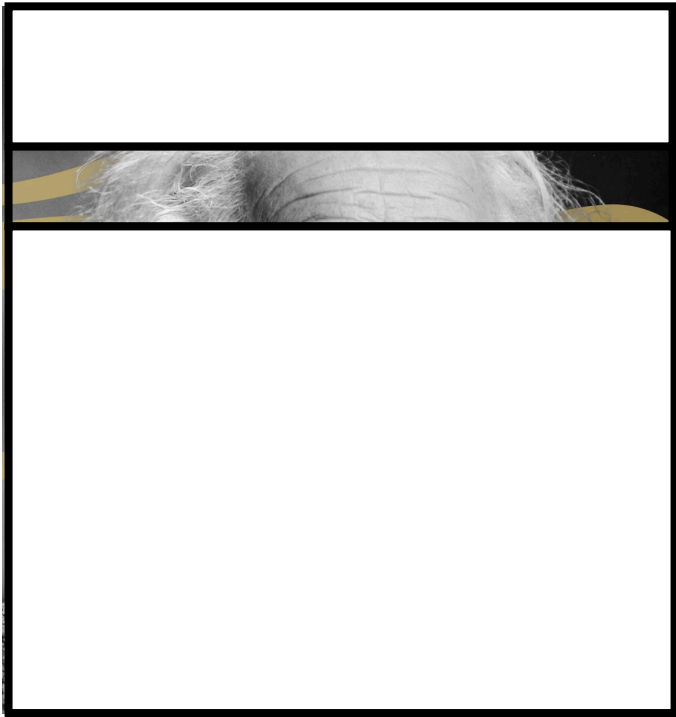
$$(t, x_t, y_t, z_t)$$



The particle seems to be moving on a plane embedded in 3D space. So we can once again get rid of a dimension here. Next, consider the following "matrix" of shape 100×100 .



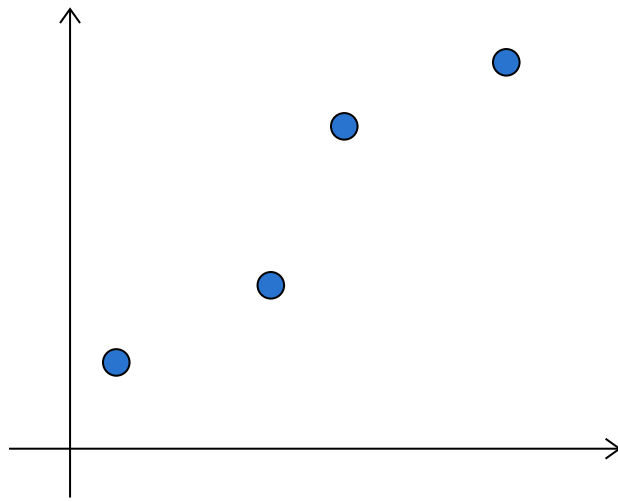
If you treat each row as a data-point, you notice that there is a lot of redundancy in the image. It is actually possible to represent the image using fewer dimensions.



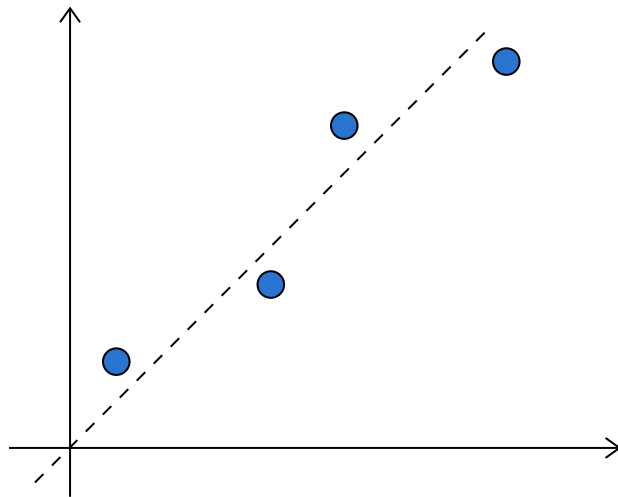
[123 ... 100 ... 10]

[123 ... 100 ... 10]

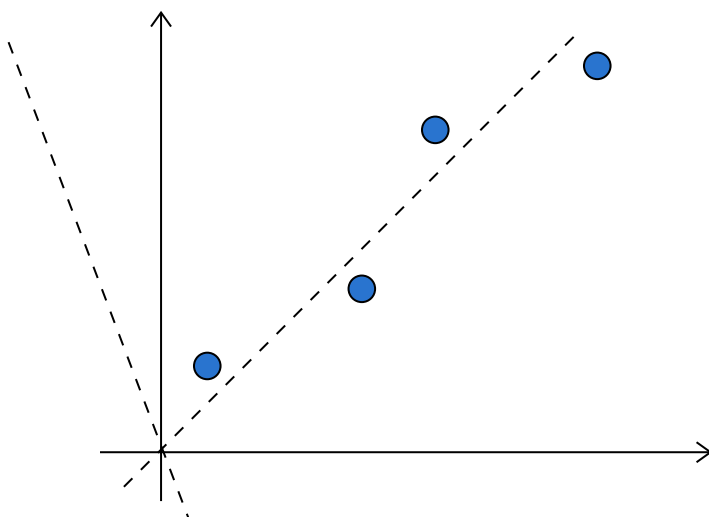
Back to the particle motion example. Notice that the sensors may not return true readings. They may be noisy. So even if the particle moves in a straight line, the actual reading may not correspond to points on a line.



Noisy measurements is a reality in any data-science problem.



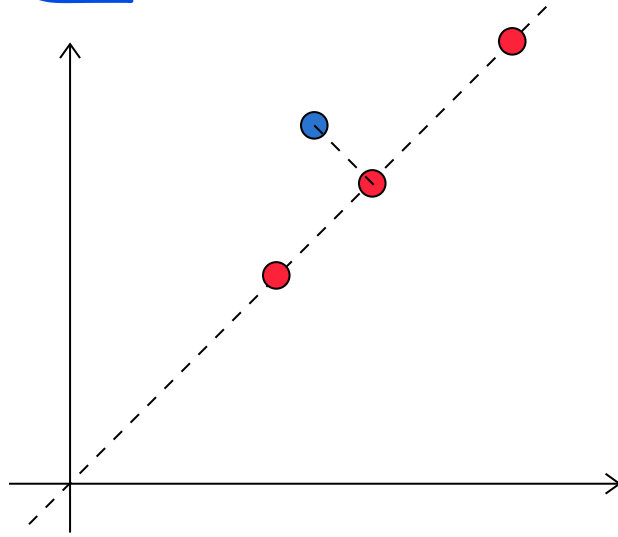
In the case of noisy measurements, how do we find a single line that will roughly correspond to the direction of motion of the particle? Of the two possible trajectories, the one "closer" to the data-points seems to be a "better fit".



2. Representatives and projections

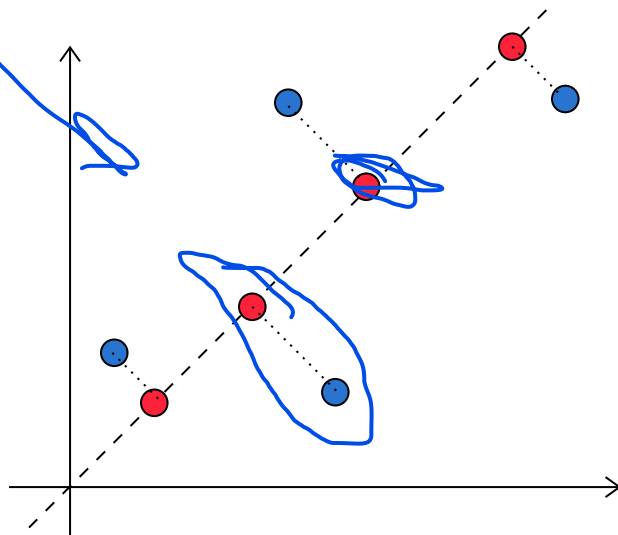
For each data-point (blue point), we associate some point on the line and

call it a representative of it.

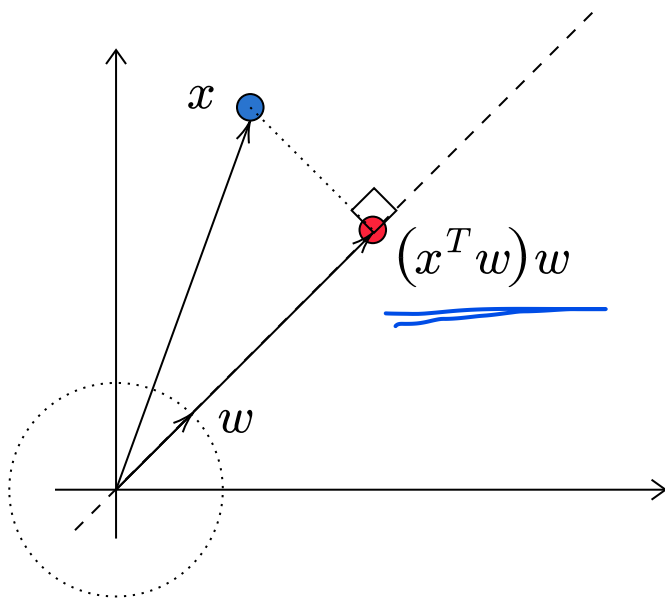


If the particle's true motion is along the line, then the red point in the middle is probably where it actually is. Alternatively, the distance of the blue point to the "true" line of motion is smallest when the red point (representative) is the projection of the blue point on the line.

The projections of the data-points are the ones shown here in red.



Next question is, how to compute the orthogonal projection of a point on a line. Recall that the goal we are after is to find the approximate line that best fits the data ("approximate direction of motion of the particle")



Let the line be represented by a unit vector w , meaning $\|w\| = 1$. The projection matrix is:

$$P = \frac{ww^T}{w^T w} = ww^T$$

The projection of a data-point x on the line passing through w is given by:

$$\begin{aligned} Px &= ww^T x \\ &= w(w^T x) \\ &= w(w^T x)^T \\ &= w(x^T w) \\ &= (x^T w)w \end{aligned}$$

The projection of the entire dataset on the line passing through the unit vector w is:

$$\begin{aligned} 1 &\rightarrow (x_1^T w)w \\ 2 &\rightarrow (x_2^T w)w \\ &\vdots \\ n &\rightarrow (x_n^T w)w \end{aligned}$$

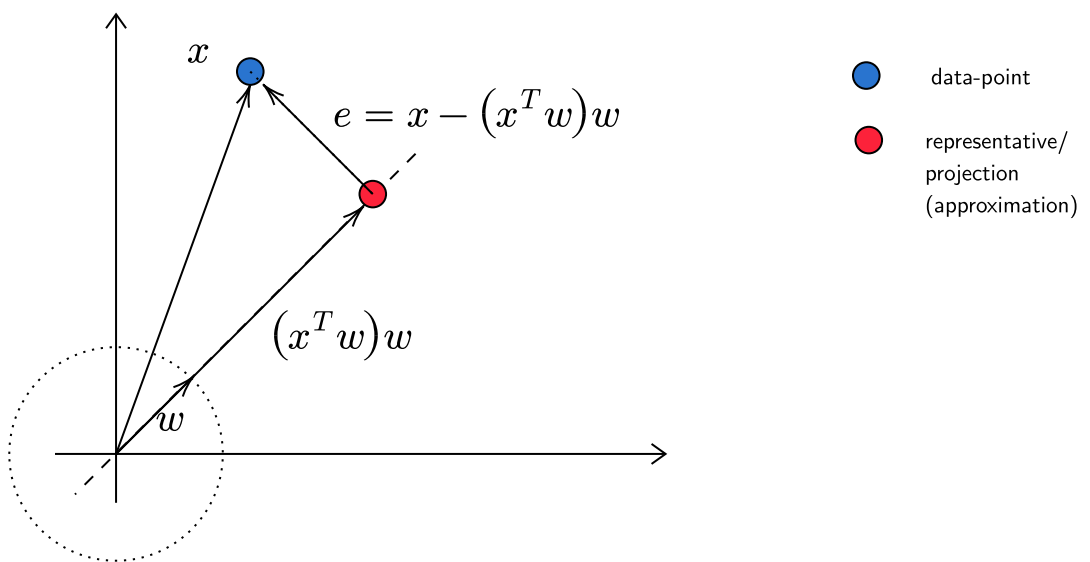
In general, for the i^{th} data-point, the projection on the line is:

$$(x_i^T w)w$$

We have to be careful with this formula. This is true only if $\|w\| = 1$.

3. Minimizing the reconstruction error

The goal is to find the line that is closest to the data. This means, the sum of the distances of all the data-points from the line has to be minimum. We have approximated the location of the blue-point with the red-point. So there is an error in approximation.



For one point x , the error in approximation is given by the error vector:

$$e = x - (x^T w)w$$

The magnitude of the error is:

$$\|e\| = \|x - (x^T w)w\|$$

It is easier to work with squared error; we will call the following quantity the reconstruction error of one data-point:

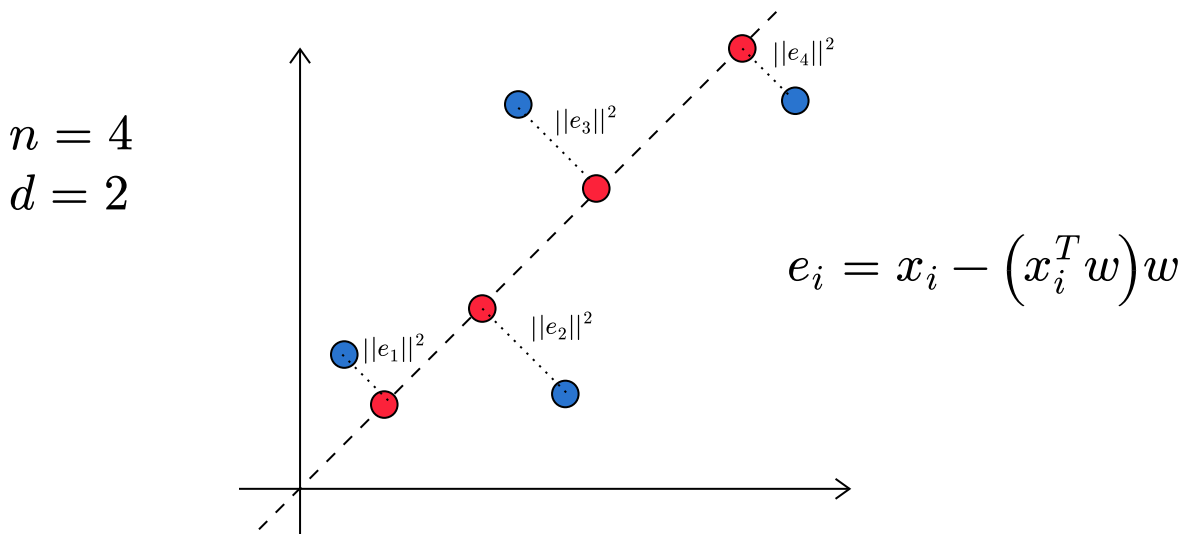
$$\|e\|^2 = \|x - (x^T w)w\|^2$$

Why is it called the reconstruction error?

- The original dataset made up of the blue points are in 2D space.
- The red points are the projections (representatives) of the blue-points

which are in a one-dimensional subspace of $\mathbb{R}^2$.

- The outcome of dimensionality reduction is to throw away the original points and retain the red points which only need one dimension, effectively.
- In this process, we are reconstructing the dataset.



The goal is to minimize the reconstruction error, which is now averaged out over all the data-points:

$$\frac{1}{n} (||e_1||^2 + \dots + ||e_n||^2)$$

Expanding this:

$$L(w) = \frac{1}{n} [||x_1 - (x_1^T w)w||^2 + \dots + ||x_n - (x_n^T w)w||^2]$$

$x_1, \dots, x_n \in \mathbb{R}^d$ are fixed once we are given the dataset. So the error is a function of w . Therefore, we can formulate the goal as minimizing the reconstruction error:

$$\min_w L(w)$$

This is an unconstrained optimization problem in w .

4. Reconstruction error in detail

Expanding this:

$$\min_w \frac{1}{n} \left[\|x_1 - (x_1^T w)w\|^2 + \dots + \|x_n - (x_n^T w)w\|^2 \right]$$

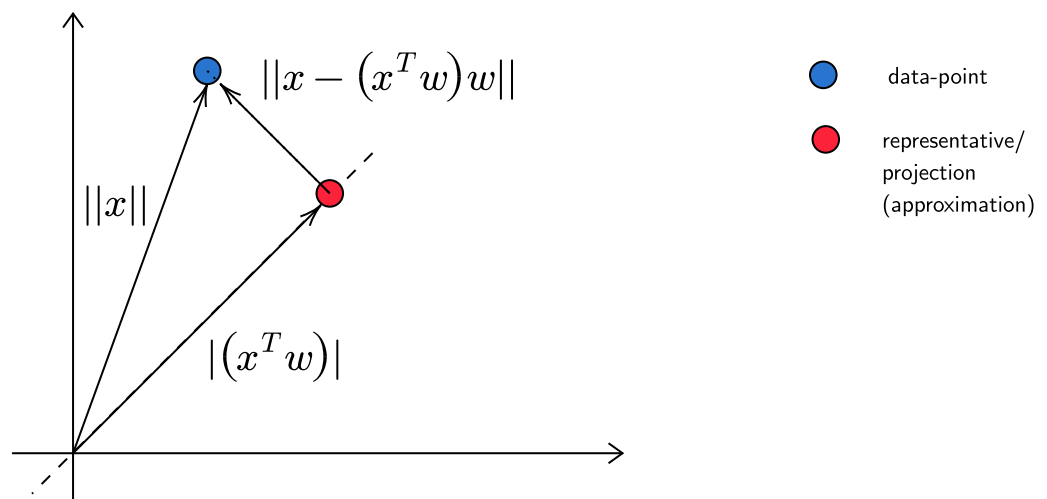
Consider one of the n terms in the sum of $L(w)$:

$$\begin{aligned} \|x_i - (x_i^T w)w\|^2 &= [x_i - (x_i^T w)w]^T [x_i - (x_i^T w)w] \\ &= x_i^T x_i - 2(x_i^T w)^2 + (x_i^T w)^2 \\ &= \|x_i\|^2 - (x_i^T w)^2 \end{aligned}$$

For the second line:

$$x_i^T (x_i^T w)w = (x_i^T w)(x_i^T w)$$

Alternatively:



From Pythagoras theorem:

$$\|x\|^2 = (x^T w)^2 + \|x - (x^T w)w\|^2$$

Rearranging:

$$\|x - (x^T w)w\|^2 = \|x\|^2 - (x^T w)^2$$

Therefore, $L(w)$:

$$L(w) = \frac{1}{n} \sum_{i=1}^n \|x_i\|^2 - (x_i^T w)^2$$

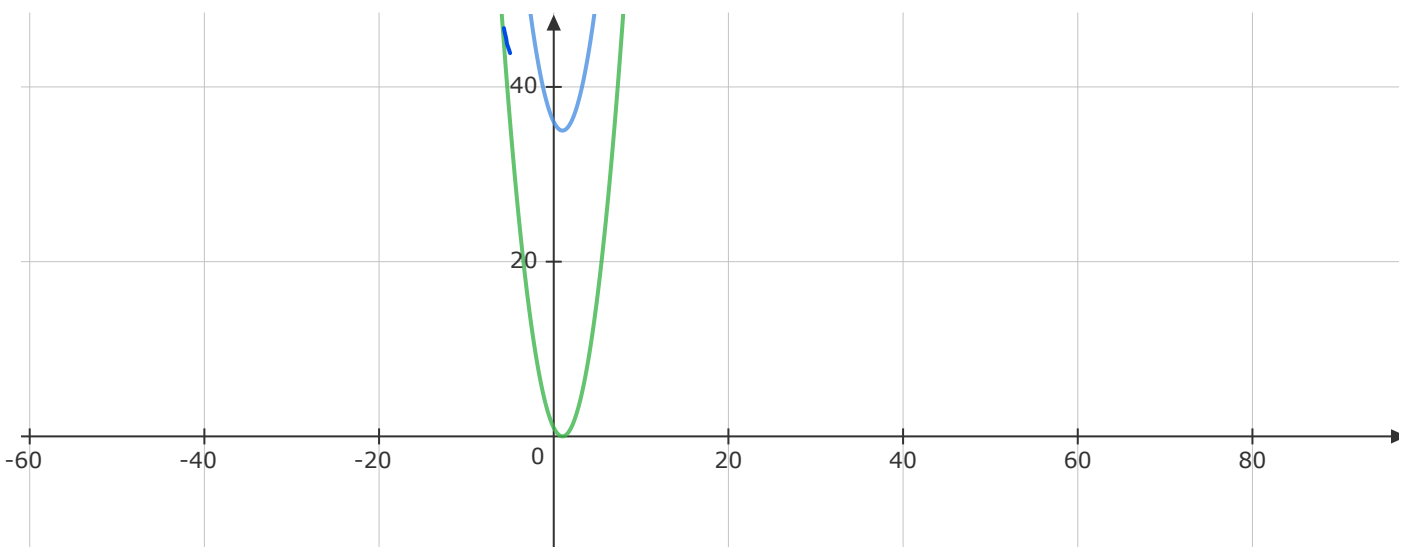
Consider the problem:

$$\text{minimize } f(a) = 35 + (a - 1)^2$$

Now, consider a related problem

$$\text{minimize } g(a) = (a - 1)^2$$

Is the solution going to be different? The a -axis where the function attains its minimum doesn't depend on the adding or subtracting of constants.



Since we are trying to minimize $L(w)$ and L depends only on w , we can get rid of the $\|x_i\|^2$ term. We can therefore drop $\|x_i\|^2$ from the

minimization problem. The equivalent optimization problem is:

$$\min_w \frac{-1}{n} \left[(x_1^T w)^2 + \dots + (x_n^T w)^2 \right]$$

Minimizing a function is the same as maximizing its negative.

$$\max_w \frac{1}{n} \left[(x_1^T w)^2 + \dots + (x_n^T w)^2 \right]$$

Let's take one term from the objective and rearrange it:

$$\begin{aligned} (x_i^T w)^2 &= (w^T x_i)(x_i^T w) \\ &= w^T (x_i x_i^T) w \end{aligned}$$

Making use of this form, we get:

$$\begin{aligned} &\frac{1}{n} \left[(x_1^T w)^2 + \dots + (x_n^T w)^2 \right] \\ &= \frac{1}{n} \left[w^T (x_1 x_1^T) w + \dots + w^T (x_n x_n^T) w \right] \\ &= \frac{1}{n} w^T \left[x_1 x_1^T + \dots + x_n x_n^T \right] w \\ &= w^T \left[\frac{1}{n} (x_1 x_1^T + \dots + x_n x_n^T) \right] w \end{aligned}$$

Now, calling the expression inside the square bracket C :

$$C = \frac{1}{n} (x_1 x_1^T + \dots + x_n x_n^T)$$

The optimization problem becomes:

$$\max_w w^T C w$$

This is a neat way of expressing the objective:

- All the information that comes from the dataset goes into C
- w sits outside separately

5. Covariance matrix

Convention: The mean of the data-points is zero.

$$\bar{x} = \frac{1}{n}(x_1 + \dots + x_n) = 0$$

Such a dataset with mean zero is called a mean-centered dataset.

For a mean-centered dataset, C is called the covariance matrix:

$$C = \frac{1}{n}(x_1 x_1^T + \dots + x_n x_n^T)$$

Recall that there are n data-points in $\mathbb{R}^d$

- $x_i \in \mathbb{R}^d$
- $x_i x_i^T$ is a matrix of shape $d \times d$ of rank 1
- C is of shape $d \times d$

Let's look at $d = 2$. Assume that you have two random variables X_1, X_2 .

$$C = \begin{bmatrix} \sigma_1^2 & \text{cov}(x_1, x_2) \\ \text{cov}(x_1, x_2) & \sigma_2^2 \end{bmatrix}$$

The diagonal elements give the variance of X_1 and X_2 .

$$\sigma_1^2 = E[(X_1 - E[X_1])^2]$$

$$\sigma_2^2 = E[(X_2 - E[X_2])^2]$$

How do X_1 and X_2 co-vary? If X_1 increases, what happens to X_2 .

$$\text{cov}(X_1, X_2) = E[(X_1 - E[X_1])(X_2 - E[X_2])]$$

In general:

$$\text{cov}(X_i, X_j) = E[(X_i - E[X_i])(X_j - E[X_j])]$$

Notice that if $i = j$, this just becomes the formula for the variance of X_i . That is the reason the diagonal elements are the individual variances.

6. Recap

We now need to look at the following optimization problems:

$$\max_{\|w\|=1} w^T C w$$

Let's look at the dataset $D = \{x_1, \dots, x_n\}$ and its relationship with the covariance matrix. We can add the data-points to a matrix, called the data-matrix.

$$X = \begin{bmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{bmatrix}$$

Some textbooks follow the convention $n \times d$. In such a case every row is a data-point.

What is the relationship between the covariance matrix and the data-matrix.

$$C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$$

$$C = \frac{1}{n} (x_1 x_1^T + \dots + x_n x_n^T) = \frac{1}{n} X X^T$$

$x y^T$ is the outer-product and it is a matrix of rank-1.

$$X X^T = \begin{bmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{bmatrix} \begin{bmatrix} - & x_1^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix}$$

$$= x_1 x_1^T + \dots + x_n x_n^T$$

Matrix

Remark: The formula $C = \frac{1}{n} X X^T$ is correct only if the means of the columns is zero. If not, $C = \frac{1}{n} (X - \mu)(X - \mu)^T$.

The relationship between the data-matrix X and the covariance matrix:

$$C = \frac{1}{n} X X^T$$

C has shape $d \times d$

7. Solving the optimization problem

$$\max_{||w||=1} w^T C w$$

For those interested: Rayleigh quotient.

Some properties of the covariance matrix:

- $C = \frac{1}{n} X X^T$
- C is square and symmetric.
 - C has shape $d \times d$
 - $C^T = C$
- By the spectral theorem, C is orthogonally diagonalizable
- C is positive semi-definite

To show the last point, let's say (λ_i, w_i) is an eigenpair:

$$\begin{aligned}
 C w_i &= \lambda_i w_i \\
 \frac{1}{n} X X^T w_i &= \lambda_i w_i \\
 \frac{1}{n} w_i^T X (X^T w_i) &= \lambda_i w_i^T w_i \\
 \frac{1}{n} (X^T w_i)^T (X^T w_i) &= \lambda_i ||w_i||^2 \\
 \frac{1}{n} ||X^T w_i||^2 &= \lambda_i ||w_i||^2
 \end{aligned}$$

We see that $\lambda_i \geq 0$. Therefore, we can list down the d eigenvalues and unit-norm eigenvectors of C as:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$$

$$w_1, w_2, \dots, w_d$$

Back to the optimization problem:

$$\max_{||w||=1} w^T C w$$

The maximum value of $w^T C w$ is achieved when $w = w_1$, which is called the first principal component of the dataset. We have found out one principal component.

$$w_1 = \arg \max_{||w||=1} w^T C w$$

argmax returns the argument at which the objective function attains its optimum.

Note on argmin

Consider

$$\min_x f(x)$$

where

$$f(x) = (x - 1)^2 + 3$$

Then:

$$\min_x f(x) = 3$$

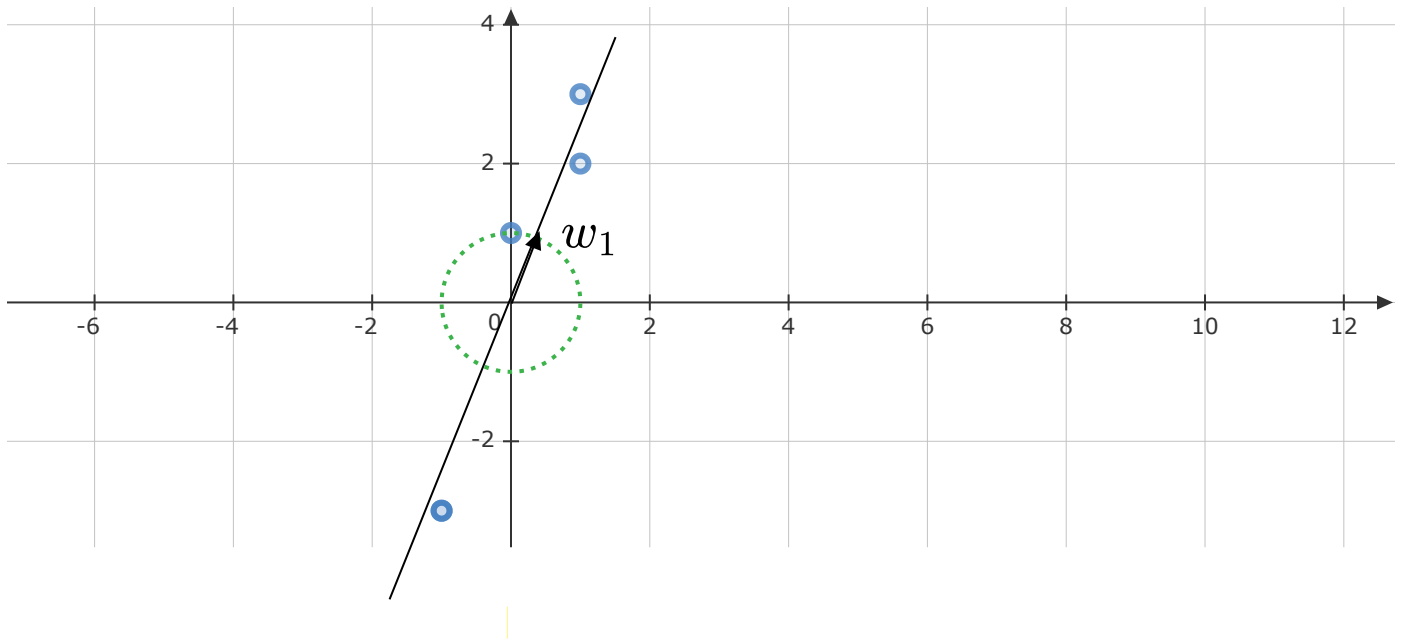
$$\arg \min_x f(x) = 1$$

The point at which f attains its minimum is the argmin.

This completes the "reconstruction error minimization" perspective.

A few points regarding the first principal component (PC):

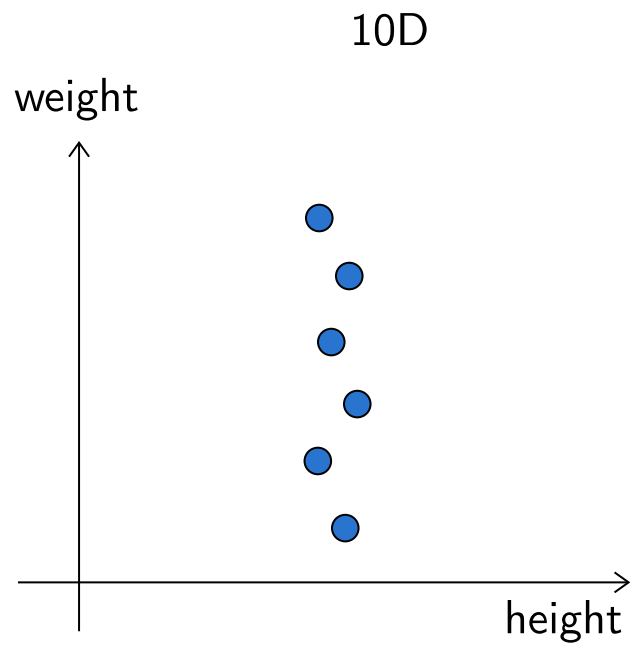
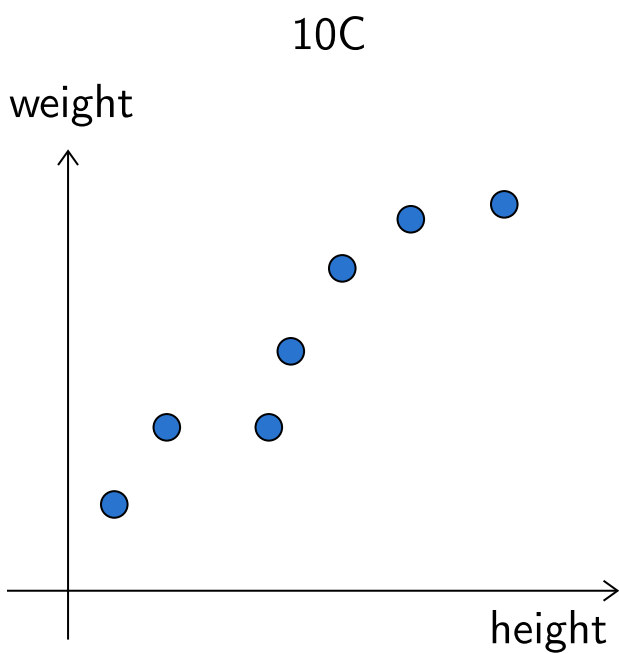
- $w_1 \in \mathbb{R}^d$
- Vector in $\mathbb{R}^d$, meaning, it is in the same space as the data-points
- w_1 is the vector such that if you project all the data-points onto w_1 , the reconstruction error will be minimum



8. Variance Maximization

Reading Number	Weight Measurement
1	60
2	60.1
...	...
10	59.9

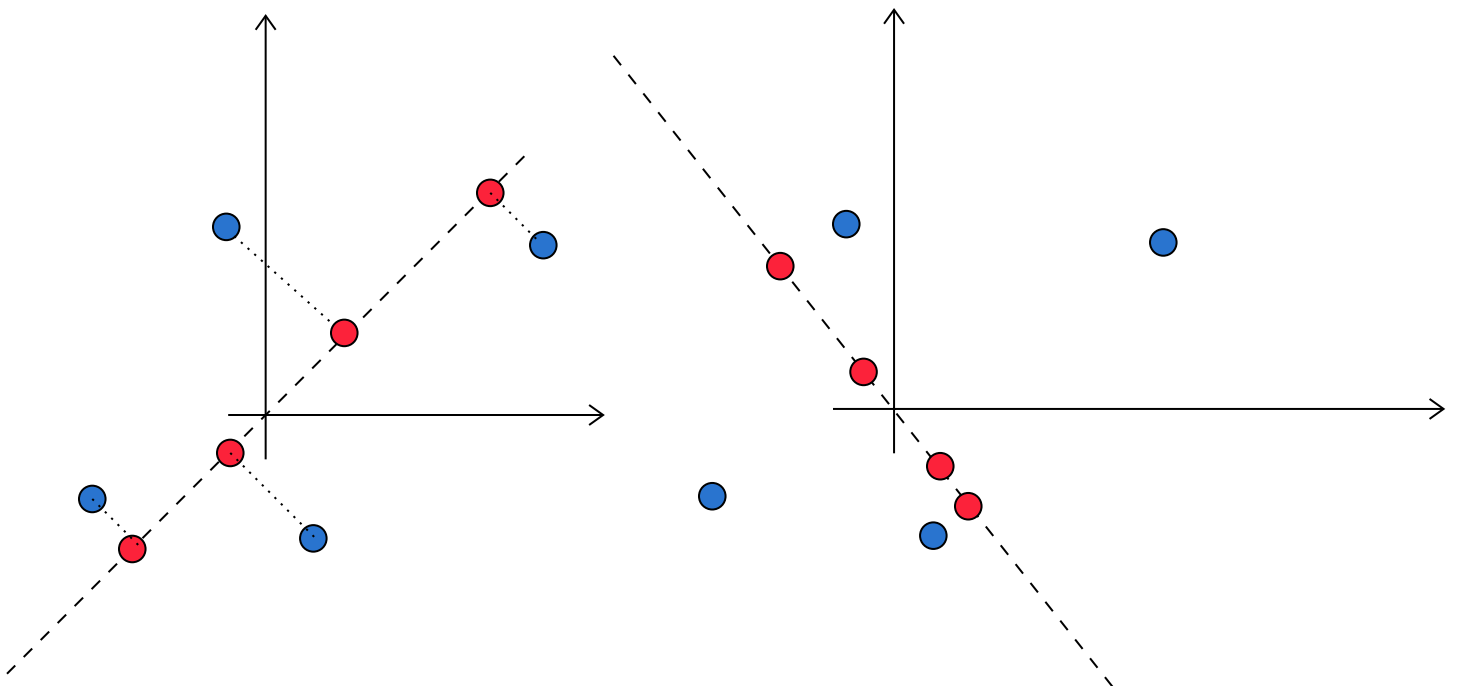
Assume that we take measurements of some physical quantity at a fixed time every day. What would we like the variance to be? We would like it to be small. One way to understand variance is to look it as a measure of error in a physical measurement. Now consider a different scenario. We have collected data of heights and weights of individuals in two classes:



The variance in the height of 10C is higher than that of 10D.

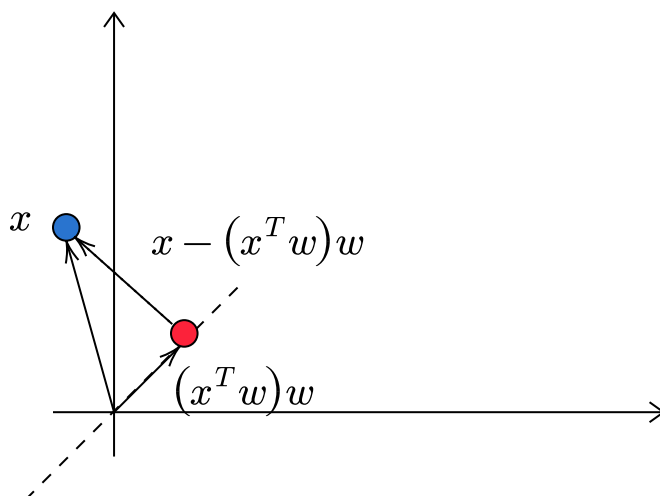
- Variance is the information content in a feature
- Greater the variance in a feature, greater the discriminative power of the feature

Goal of data-science is extract insights from data so that we can make decisions based on that. Data science is the process of converting data to decisions. We now need to find the direction in which the variance (information) is maximum.



The best direction could be one which maximizes the variance of the dataset along it. The main insight is that the direction which minimizes the reconstruction error is exactly the same direction which maximizes the

variance. Next, we find the variance of a set of points along a line w .

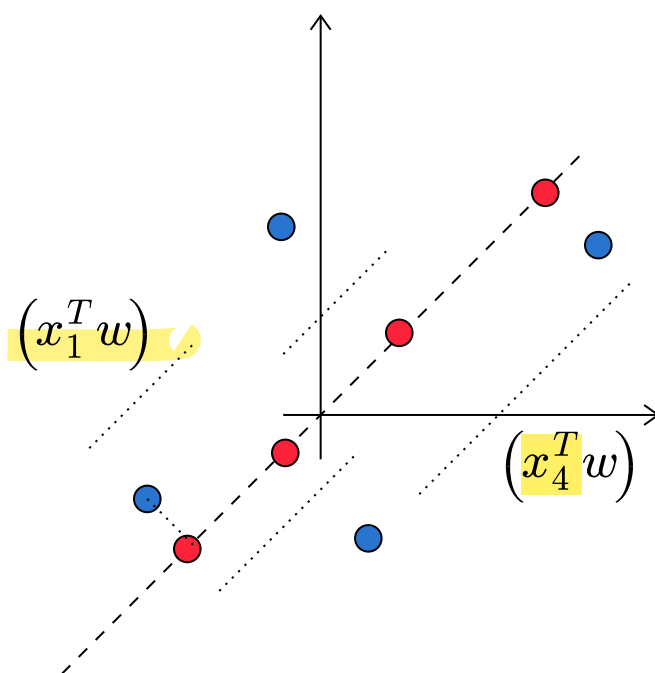


The squared distance of the projected point from the origin is:

$$(x^T w)^2$$

If we have n points, the sum of squared distances is the variance of the projected points. This is because the mean of the projected points is also $(0, 0)$.

$$\text{var}(D, w) = \frac{1}{n} \left[(x_1^T w)^2 + \dots + (x_n^T w)^2 \right]$$



Let's now rearrange this expression a bit:

$$\begin{aligned}
 \text{var}(D, w) &= \frac{1}{n} \left[(x_1^T w)^2 + \dots + (x_n^T w)^2 \right] \\
 &= w^T \left(\frac{1}{n} [x_1 x_1^T + \dots + x_n x_n^T] \right) w \\
 &= w^T C w
 \end{aligned}$$

This is the same objective function as before!

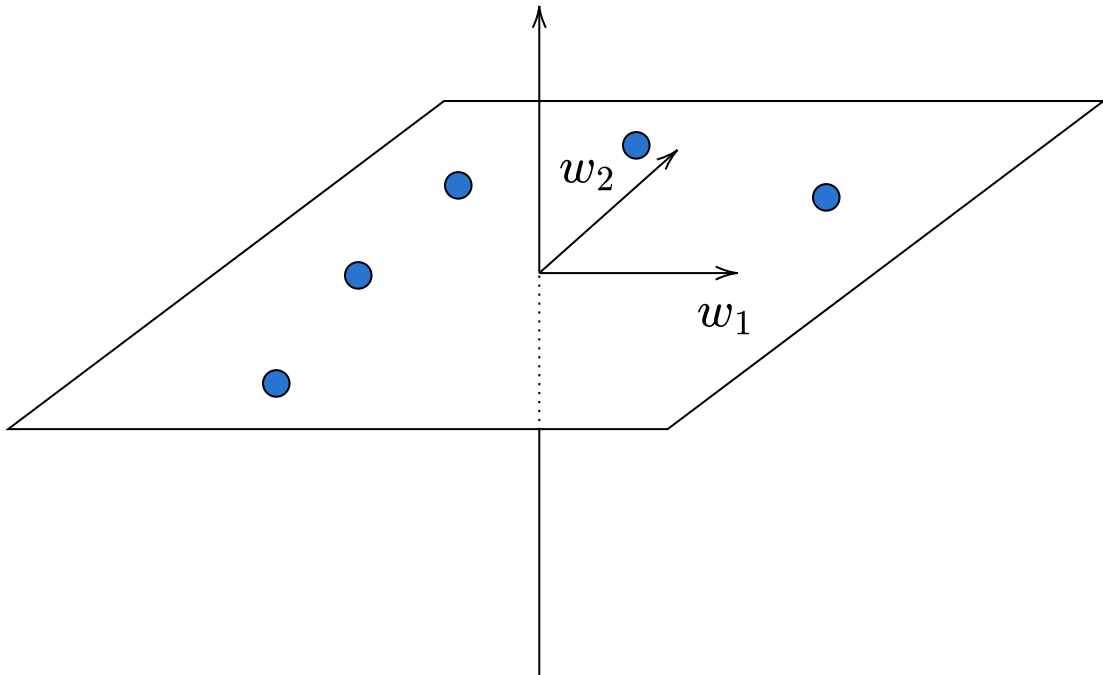
$$w_1 = \arg \max_{\|w\|=1} w^T C w$$

The direction along which the reconstruction error is minimized is the same as the direction along which the variance is maximized.

$$\begin{aligned}
 w_1^T C w_1 &= w_1^T (\lambda_1 w_1) \\
 &= \lambda_1 (w_1^T w_1) \\
 &= \lambda_1
 \end{aligned}$$

The value $w_1^T C w_1$ is the variance of the dataset along the first principal component and this is nothing but the largest eigenvalue of C . In general, $w^T C w$ is the variance of the dataset along the direction w . The largest eigenvalue of the covariance matrix is the variance of the dataset along the corresponding eigenvector, which is called the first principal component.

9. Other PCs



The eigenvectors of C are called the principal components.

- $\lambda_1 \geq \dots \geq \lambda_d$
- $w_1, \dots, w_d$

If we want to represent a data-point using only the first PC:

$$x_i \rightarrow (x_i^T w_1) w_1$$

If there are two PCs:

$$x_i \rightarrow (x_i^T w_1) w_1 + (x_i^T w_2) w_2$$

In general, if we have k PCs:

$$x_i \rightarrow (x_i^T w_1) w_1 + \dots + (x_i^T w_k) w_k$$

$$x_i' = (x_i^T w_1) w_1 + \dots + (x_i^T w_k) w_k$$

$x_i' \in \mathbb{R}^d$. If we want to throw away certain dimensions, we only retain the scalar projections along the first k directions.

$$x_i \rightarrow \begin{bmatrix} x_i^T w_1 \\ \vdots \\ x_i^T w_k \end{bmatrix} \in \mathbb{R}^k$$


- First, replace x_i with its projection on the k PCs
- Retain only the scalar projections on the top k PCs

10. Reference

- [Prof Arun Rajkumar's MLT lectures for week-1 starting from the PCA part.](#)
- [A tutorial on PCA, J Shlens](#)